

## 1) собираем

```
configure --prefix=$PWD/_install --static
gadirov@nail-gad:~/prog/kernel/CROSS_COMPILE/zlib-1.3.2$ ./configure --prefix=$PWD/_install --static
Checking for gcc...
Building static library libz.a version 1.3.2 with gcc.
Checking for size_t... Yes.
Checking for off64_t... Yes.
Checking for fseeko... Yes.
Checking for strerror... Yes.
Checking for unistd.h... Yes.
Checking for stdarg.h... Yes.
Checking whether to use vs[n]printf() or s[n]printf()... using vs[n]printf().
Checking for vsnprintf() in stdio.h... Yes.
Checking for return value of vsnprintf()... Yes.
Checking for attribute(visibility) support... Yes.
Checking for s390x build ... No.
gadirov@nail-gad:~/prog/kernel/CROSS_COMPILE/zlib-1.3.2$
```

## 2) собираем под ARM и устанавливаем

```
rm -f *.gda *.gcnv *.gcov
rm -f contrib/inffast9/*.gda contrib/inffast9/*.gcnv contrib/inffast9/*.gcov
gadirov@nail-gad:~/prog/kernel/CROSS_COMPILE/zlib-1.3.2$ make CC=arm-linux-gnueabi-gcc
arm-linux-gnueabi-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -I. -c -o example.o test/example.c
arm-linux-gnueabi-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o adler32.o adler32.c
arm-linux-gnueabi-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o crc32.o crc32.c
arm-linux-gnueabi-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o deflate.o deflate.c
arm-linux-gnueabi-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o inffast.o inffast.c
arm-linux-gnueabi-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o inflate.o inflate.c
arm-linux-gnueabi-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o inftrees.o inftrees.c
arm-linux-gnueabi-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o gzlib.o gzlib.c
arm-linux-gnueabi-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o gzread.o gzread.c
arm-linux-gnueabi-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o gzwrite.o gzwrite.c
ar rc libz.a adler32.o crc32.o deflate.o inffast.o inflate.o inftrees.o trees.o zutil.o compress.o uncompress.o gzclose.o gzlib.o gzread.o gzwrite.o
arm-linux-gnueabi-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o example example.o -L. libz.a
arm-linux-gnueabi-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -I. -c -o minigzip.o test/minigzip.c
arm-linux-gnueabi-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o minigzip minigzip.o -L. libz.a
arm-linux-gnueabi-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -I. -D_FILE_OFFSET_BITS=64 -c -o example64.o test/example.c
arm-linux-gnueabi-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o example64 example64.o -L. libz.a
arm-linux-gnueabi-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -I. -D_FILE_OFFSET_BITS=64 -c -o minigzip64.o test/minigzip.c
arm-linux-gnueabi-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o minigzip64 minigzip64.o -L. libz.a
gadirov@nail-gad:~/prog/kernel/CROSS_COMPILE/zlib-1.3.2$ file trees.o
trees.o: ELF 32-bit LSB relocatable, ARM, EABI5 version 1 (SYSV), not stripped
gadirov@nail-gad:~/prog/kernel/CROSS_COMPILE/zlib-1.3.2$
```

### 3) install

```
gadirov@nail-gad:~/prog/kernel/CROSS_COMPILE/zlib-1.3.2$ _in
_included_ssh_config_files _init_completion _installed_modules
gadirov@nail-gad:~/prog/kernel/CROSS_COMPILE/zlib-1.3.2$ make install
rm -f /home/gadirov/prog/kernel/CROSS_COMPILE/zlib-1.3.2/_install/lib/libz.a
cp libz.a /home/gadirov/prog/kernel/CROSS_COMPILE/zlib-1.3.2/_install/lib
chmod 644 /home/gadirov/prog/kernel/CROSS_COMPILE/zlib-1.3.2/_install/lib/libz.a
rm -f /home/gadirov/prog/kernel/CROSS_COMPILE/zlib-1.3.2/_install/share/man/man3/zlib.3
cp zlib.3 /home/gadirov/prog/kernel/CROSS_COMPILE/zlib-1.3.2/_install/share/man/man3
chmod 644 /home/gadirov/prog/kernel/CROSS_COMPILE/zlib-1.3.2/_install/share/man/man3/zlib.3
rm -f /home/gadirov/prog/kernel/CROSS_COMPILE/zlib-1.3.2/_install/lib/pkgconfig/zlib.pc
cp zlib.pc /home/gadirov/prog/kernel/CROSS_COMPILE/zlib-1.3.2/_install/lib/pkgconfig
chmod 644 /home/gadirov/prog/kernel/CROSS_COMPILE/zlib-1.3.2/_install/lib/pkgconfig/zlib.pc
rm -f /home/gadirov/prog/kernel/CROSS_COMPILE/zlib-1.3.2/_install/include/zlib.h /home/gadi
rov/prog/kernel/CROSS_COMPILE/zlib-1.3.2/_install/include/zconf.h
cp zlib.h zconf.h /home/gadirov/prog/kernel/CROSS_COMPILE/zlib-1.3.2/_install/include
chmod 644 /home/gadirov/prog/kernel/CROSS_COMPILE/zlib-1.3.2/_install/include/zlib.h /home/
gadirov/prog/kernel/CROSS_COMPILE/zlib-1.3.2/_install/include/zconf.h
gadirov@nail-gad:~/prog/kernel/CROSS_COMPILE/zlib-1.3.2$
```

### 4)запускаем конфигурацию

```
gadirov@nail-gad:~/prog/kernel/CROSS_COMPILE/openssh-portable$ ./configure --host=arm-linux
-gnueabihf --with-zlib=$PWD/../zlib-1.3.2/_install --prefix=$PWD/_install
checking for arm-linux-gnueabihf-cc... no
checking for arm-linux-gnueabihf-gcc... arm-linux-gnueabihf-gcc
checking whether the C compiler works... yes
checking for C compiler default output file name... a.out
checking for suffix of executables...
checking whether we are cross compiling... yes
checking for suffix of object files... o
checking whether the compiler supports GNU C... yes
checking whether arm-linux-gnueabihf-gcc accepts -g... yes
checking for arm-linux-gnueabihf-gcc option to enable C11 features... none needed
checking if arm-linux-gnueabihf-gcc supports C99-style variadic macros... yes
checking build system type... x86_64-pc-linux-gnu
checking host system type... arm-unknown-linux-gnueabihf
checking for stdio.h... yes
checking for stdlib.h... yes
checking for string.h... yes
```

```

checking compiler and flags for sanity... configure: WARNING: cross compiling: not checking
  compiler sanity
checking for setsockopt... yes
checking for dirname... yes
checking for libgen.h... (cached) yes
checking for getsppam... yes
checking for library containing basename... none required
checking for sqrt... no
checking for sqrt in -lm... yes
checking for zlib... yes
checking for zlib.h... yes
checking for deflate in -lz... yes
checking for possibly buggy zlib... configure: WARNING: cross compiling: not checking zlib
version
checking for strcasecmp... yes
checking for utimes... yes
checking for bsd/libutil.h... no
checking for libutil.h... no
checking for daemon... yes
checking for getpagesize... yes
checking whether snprintf correctly terminates long strings... configure: WARNING: cross co
mpiling: Assuming working snprintf()
checking whether snprintf understands %zu... configure: WARNING: cross compiling: Assuming
working snprintf()
checking whether vsnprintf returns correct values on overflow... configure: WARNING: cross
compiling: Assuming working vsnprintf()
checking whether snprintf can declare const char *fmt... yes
checking whether system supports SO_PEERCREC getsockopt... yes
checking if openpty correctly handles controlling tty... cross-compiling, assuming yes
checking whether AI_NUMERICSERV is declared... yes
checking if SA_RESTARTed signals interrupt select()... configure: WARNING: cross compiling:
  assuming yes
checking for getpgrp... yes
checking if getpgrp accepts zero args... yes
checking for openssl... /usr/bin/openssl
configure: error: *** working libcrypto not found, check config.log

```

## 5) настройка в openssl

```

gadirov@nail-gad:~/prog/kernel/CROSS_COMPILE/openssl-3.0.15$ ./Configure linux-armv4 --pref
ix=$PWD/install --cross-compile-prefix=arm-linux-gnueabihf- no-asm --static no-shared no-ds
o
Configuring OpenSSL version 3.0.15 for target linux-armv4
Using os-specific seed configuration
Created configdata.pm
Running configdata.pm
Created Makefile.in
Created Makefile
Created include/openssl/configuration.h

*****
***                                     ***
***   OpenSSL has been successfully configured                               ***
***                                     ***
***   If you encounter a problem while building, please open an           ***
***   issue on GitHub <https://github.com/openssl/openssl/issues>         ***
***   and include the output from the following command:                   ***
***                                     ***
***       perl configdata.pm --dump                                         ***
***                                     ***
***   (If you are new to OpenSSL, you might want to consult the           ***
***   'Troubleshooting' section in the INSTALL.md file first)             ***
***                                     ***
*****

```

## 6) сборка ARM

```
drwxrwxr-x 2 gadirov gadirov 4096 Sep 3 2024 tools
drwxrwxr-x 3 gadirov gadirov 4096 Sep 3 2024 util
-rw-rw-r-- 1 gadirov gadirov 100 Sep 3 2024 VERSION.dat
drwxrwxr-x 2 gadirov gadirov 4096 Sep 3 2024 VMS
drwxrwxr-x 2 gadirov gadirov 4096 Sep 3 2024 wycheproof
gadirov@nail-gad:~/prog/kernel/CROSS_COMPILE/openssl-3.0.15$ make CC=arm-linux-gnueabi-gcc AR=arm-linux-gnueabi-ar RANLIB=arm-linux-gnueabi-ranlib
/usr/bin/perl "-I." -Mconfigdata "util/dofile.pl" "-oMakefile" include/crypto/bn_conf.h.in
> include/crypto/bn_conf.h
/usr/bin/perl "-I." -Mconfigdata "util/dofile.pl" "-oMakefile" include/crypto/dso_conf.h.in
> include/crypto/dso_conf.h
/usr/bin/perl "-I." -Mconfigdata "util/dofile.pl" "-oMakefile" include/openssl/asn1.h.in > i
include/openssl/asn1.h
/usr/bin/perl "-I." -Mconfigdata "util/dofile.pl" "-oMakefile" include/openssl/asn1t.h.in >
include/openssl/asn1t.h
/usr/bin/perl "-I." -Mconfigdata "util/dofile.pl" "-oMakefile" include/openssl/bio.h.in > i
include/openssl/bio.h
/usr/bin/perl "-I." -Mconfigdata "util/dofile.pl" "-oMakefile" include/openssl/cmp.h.in > i
include/openssl/cmp.h
/usr/bin/perl "-I." -Mconfigdata "util/dofile.pl" "-oMakefile" include/openssl/cms.h.in > i
include/openssl/cms.h
/usr/bin/perl "-I." -Mconfigdata "util/dofile.pl" "-oMakefile" include/openssl/conf.h.in > i
include/openssl/conf.h
/usr/bin/perl "-I." -Mconfigdata "util/dofile.pl" "-oMakefile" include/openssl/crmf.h.in > i
include/openssl/crmf.h
```

```
use the shared libraries from the glibc version used for linking
rm -f "apps/CA.pl"
/usr/bin/perl "-I." -Mconfigdata "util/dofile.pl" \
  "-oMakefile" apps/CA.pl.in > "apps/CA.pl"
chmod a+x apps/CA.pl
rm -f "apps/tsget.pl"
/usr/bin/perl "-I." -Mconfigdata "util/dofile.pl" \
  "-oMakefile" apps/tsget.in > "apps/tsget.pl"
chmod a+x apps/tsget.pl
rm -f "tools/c_rehash"
/usr/bin/perl "-I." -Mconfigdata "util/dofile.pl" \
  "-oMakefile" tools/c_rehash.in > "tools/c_rehash"
chmod a+x tools/c_rehash
rm -f "util/shlib_wrap.sh"
/usr/bin/perl "-I." -Mconfigdata "util/dofile.pl" \
  "-oMakefile" util/shlib_wrap.sh.in > "util/shlib_wrap.sh"
chmod a+x util/shlib_wrap.sh
rm -f "util/wrap.pl"
/usr/bin/perl "-I." -Mconfigdata "util/dofile.pl" \
  "-oMakefile" util/wrap.pl.in > "util/wrap.pl"
chmod a+x util/wrap.pl
make[1]: Leaving directory '/home/gadirov/prog/kernel/CROSS_COMPILE/openssl-3.0.15'
gadirov@nail-gad:~/prog/kernel/CROSS_COMPILE/openssl-3.0.15$ file ./crypto/libcrypto-lib-context.o
./crypto/libcrypto-lib-context.o: ELF 32-bit LSB relocatable, ARM, EABI5 version 1 (SYSV), not stripped
gadirov@nail-gad:~/prog/kernel/CROSS_COMPILE/openssl-3.0.15$
```



9) сборка bin, lib

10) переносим

```
gadirov@nail-gad:~/Desktop/HW_19/rootfs$ cd bin/
gadirov@nail-gad:~/Desktop/HW_19/rootfs/bin$ ls -la --sort time
итого 26908
-rwxrwxr-x 1 gadirov gadirov 4838692 Jun 18 12:34 ssh-keyscan
drwxrwxr-x 2 gadirov gadirov 4096 Jun 18 12:34 .
-rwxrwxr-x 1 gadirov gadirov 4338536 Jun 18 12:34 ssh-agent
-rwxrwxr-x 1 gadirov gadirov 4480868 Jun 18 12:34 ssh-add
-rwxrwxr-x 1 gadirov gadirov 4805536 Jun 18 12:34 ssh-keygen
-rwxrwxr-x 1 gadirov gadirov 746408 Jun 18 12:34 sftp
-rwxrwxr-x 1 gadirov gadirov 711436 Jun 18 12:33 scp
-rwxrwxr-x 1 gadirov gadirov 6070560 Jun 18 12:33 ssh
drwxrwxr-x 5 gadirov gadirov 4096 Jun 18 12:25 ..
lrwxrwxrwx 1 gadirov gadirov 7 Jun 18 06:03 arch -> busybox
lrwxrwxrwx 1 gadirov gadirov 7 Jun 18 06:03 ash -> busybox
lrwxrwxrwx 1 gadirov gadirov 7 Jun 18 06:03 base32 -> busybox
lrwxrwxrwx 1 gadirov gadirov 7 Jun 18 06:03 base64 -> busybox
```

11) настроили

```
No user exists for uid 0
~ # ssh -V
No user exists for uid 0
~ # echo "root:x:0:0:root:/root:/bin/sh" > /etc/passwd
~ # echo "root::0:0:99999:7:::" > /etc/shadow
~ # echo "root:x:0:" > /etc/group
~ # ssh -V
OpenSSH_10.3p1, OpenSSL 3.0.15 3 Sep 2024
~ #
```